

Week 4 – DATA VISUALIZATION & CAPSTONE PROJECT

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Course Name – Data Science Using Python

Data Visualization & Capstone Project

This report presents data cleaning, exploratory data analysis, sales prediction using regression, Excel analysis, Power BI dashboard requirements, findings and recommendations for the provided sales dataset.

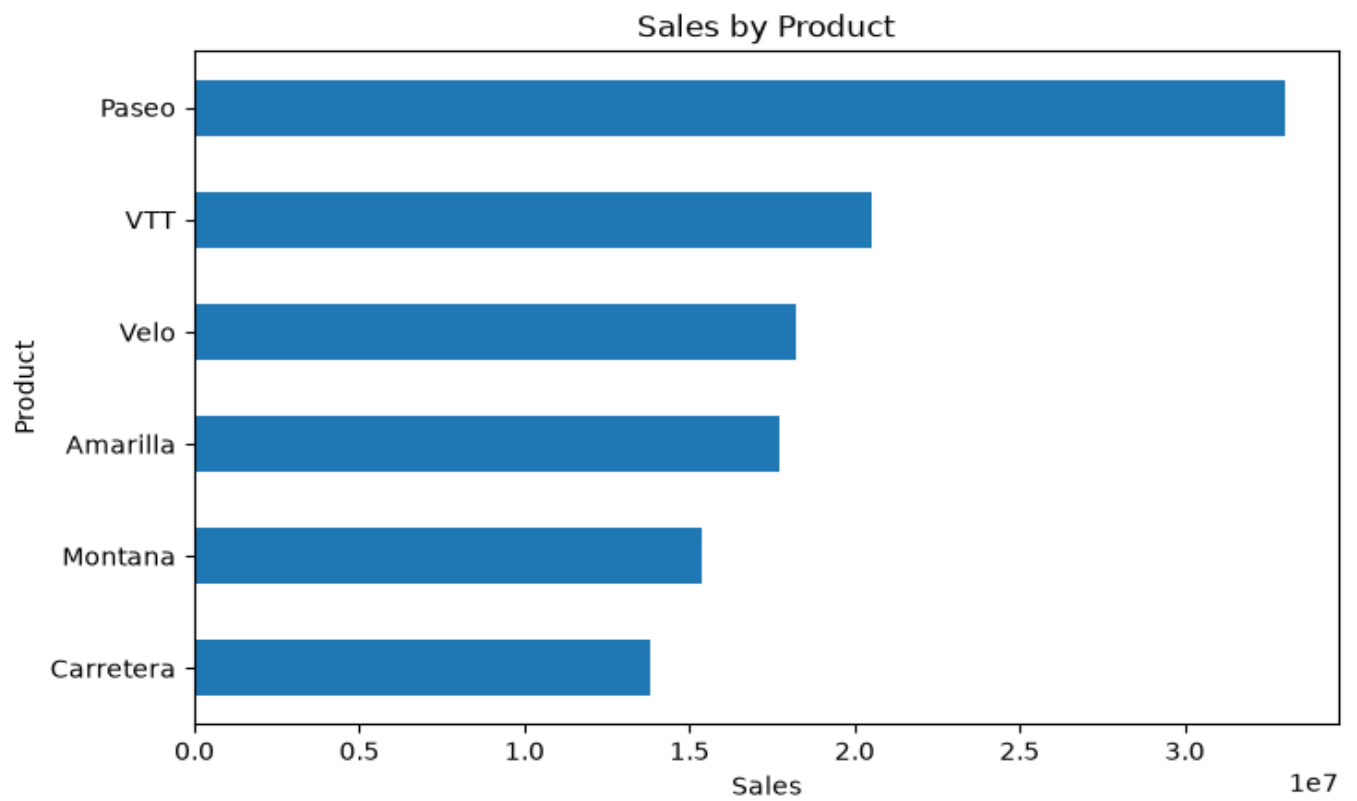
1. Dataset Overview

The dataset contains 700 records and 16 columns. The dates range from September 2013 to December 2014. There are no duplicate rows after cleaning. The Discount Band column had 53 missing values, which were treated as 'Unknown'.

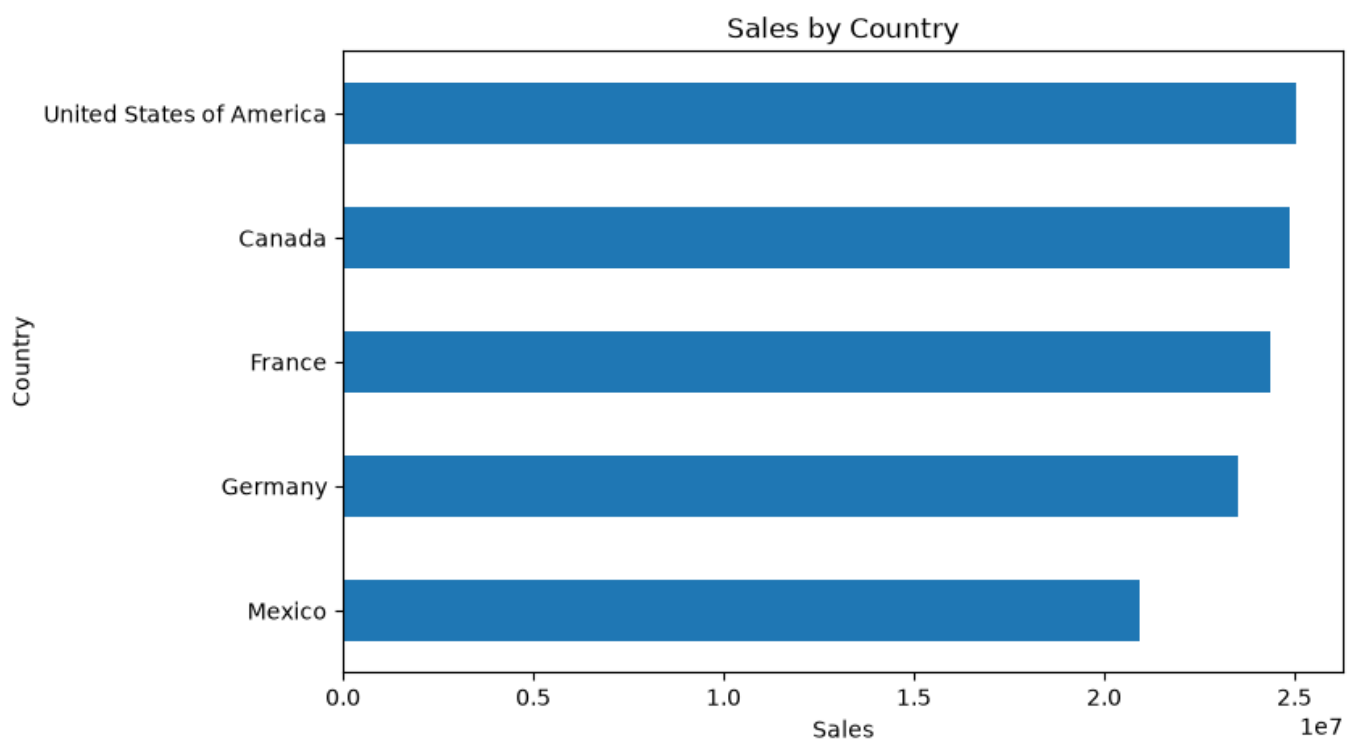
2. Key Metrics

Metric	Value
Total Sales	118,726,350.26
Total Profit	16,893,702.26
Total Units Sold	1,125,806
Gross Sales	127,931,598.50
Discounts	9,205,248.24
Overall Profit Margin	14.23%

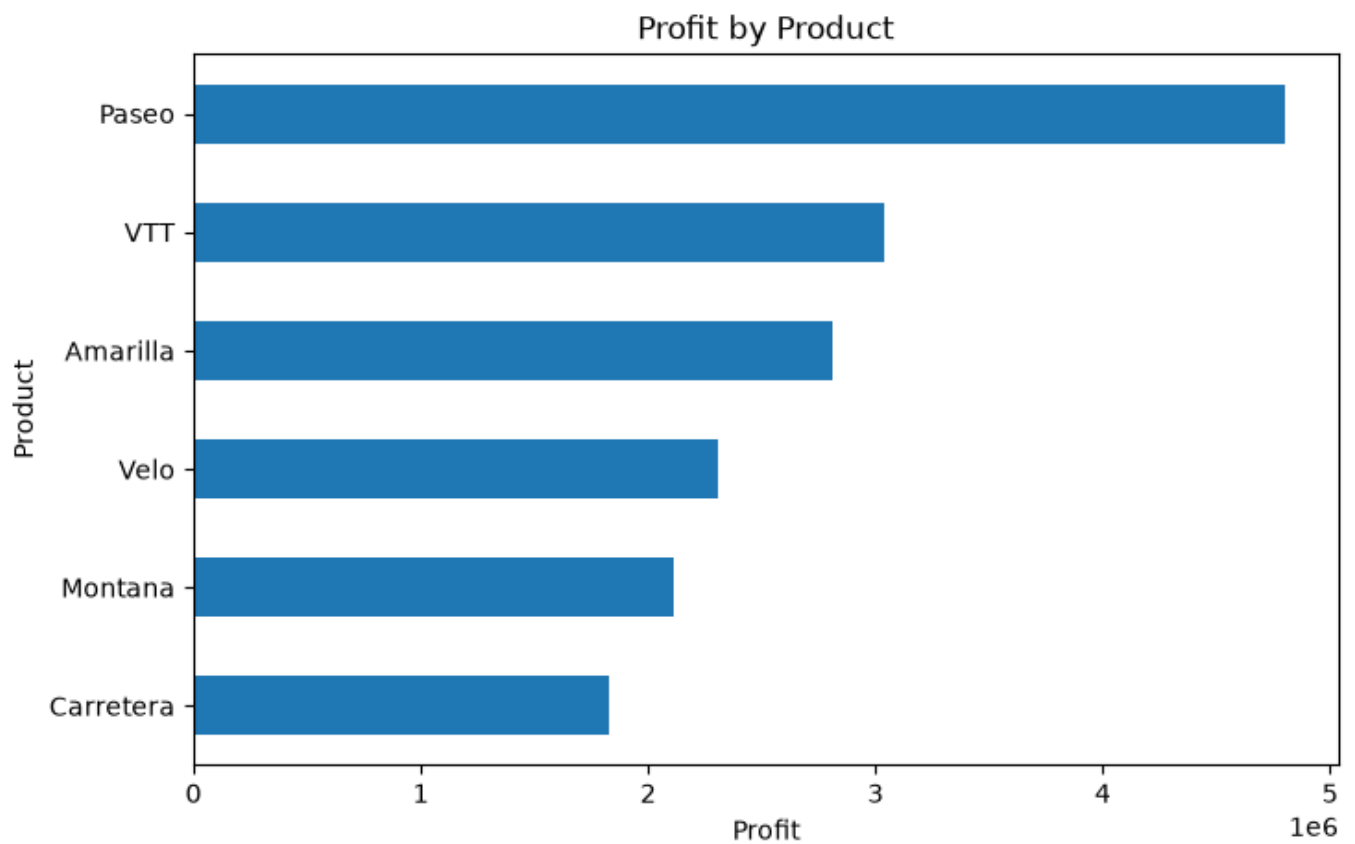
3. Exploratory Data Analysis



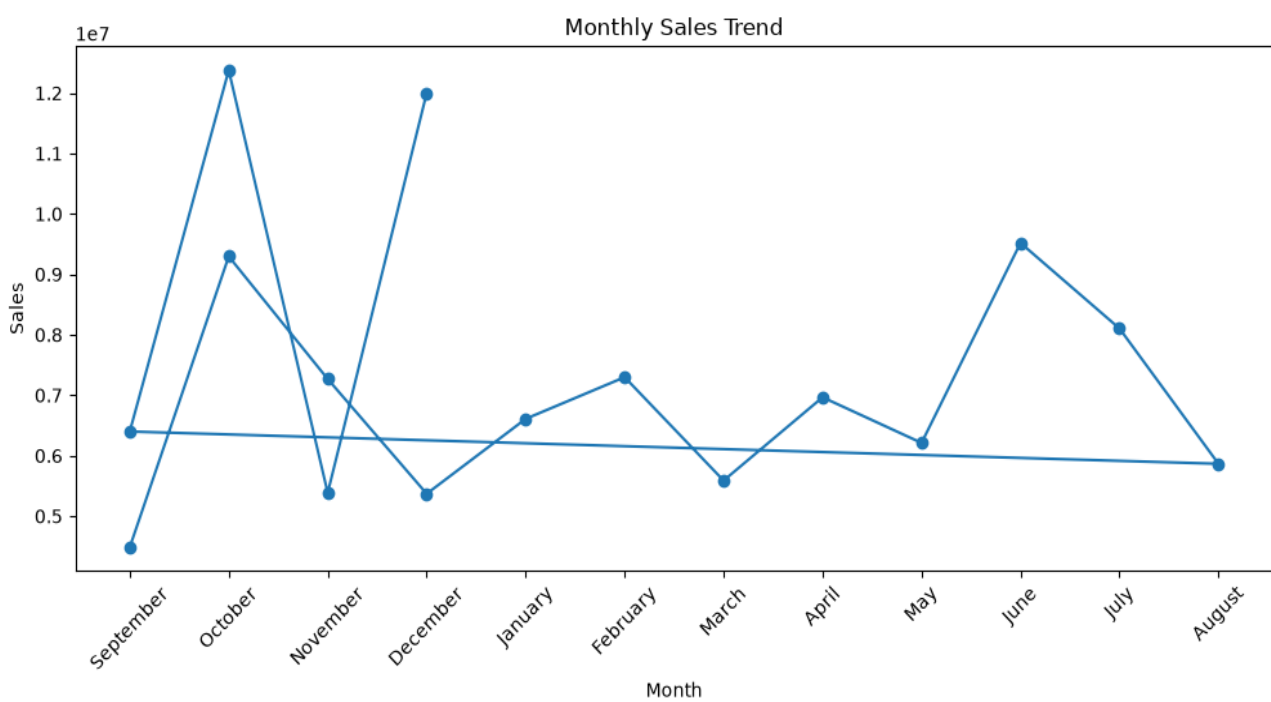
Sales by Product



Sales by Country



Profit by product

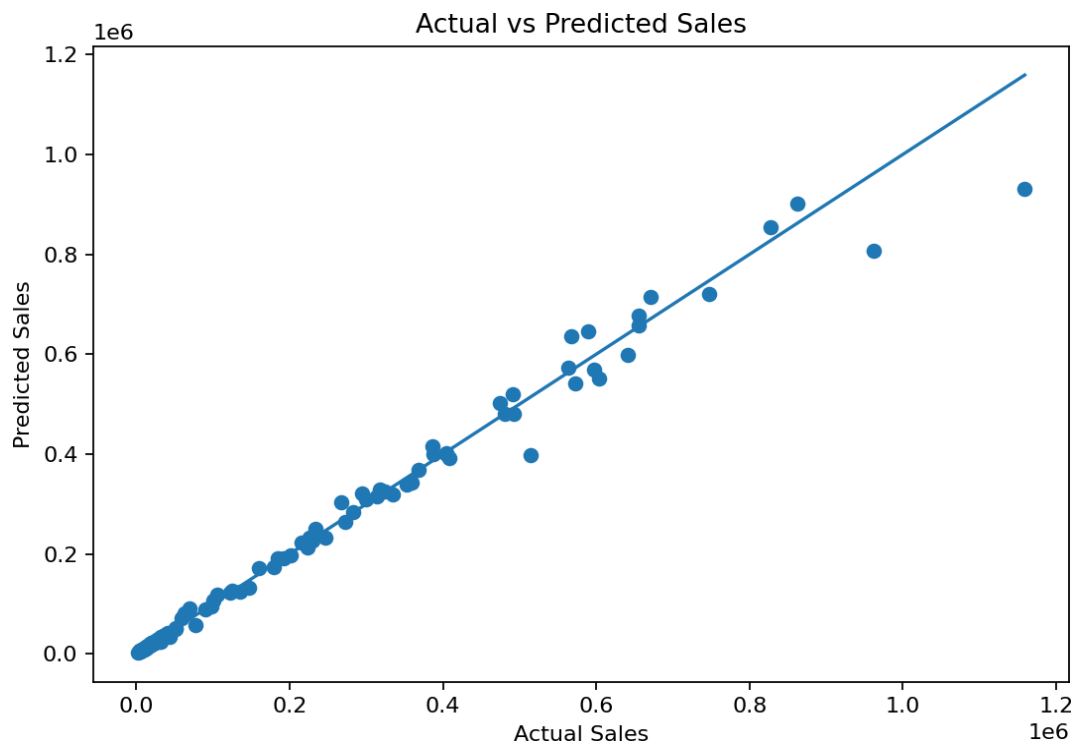


Monthly Sales Tr

4. Regression Model

A Random Forest Regression model was used to predict Sales. The input variables included segment, country, product, discount band, units sold, manufacturing price, sale price, month number and year. Gross Sales, Discounts, COGS and Profit were not used as predictors because they are directly related to the Sales calculation and could cause target leakage.

Metric	Result
Model	Random Forest Regressor
Training Records	560
Testing Records	140
MAE	10,904.44
RMSE	29,112.95
R ² Score	0.9847



Actual versus predicted sales on the test set.

5. Key Findings

- Total sales are 118.73 million and total profit is 16.89 million.
- Paseo is the highest-sales product with 33.01 million in sales.
- Government is the largest sales segment with 52.50 million in sales.
- Enterprise records negative total profit of 614,545.62 in the dataset.
- 2014 has substantially higher sales than 2013.
- The low discount band has a higher profit margin than the high discount band.

Recommendations

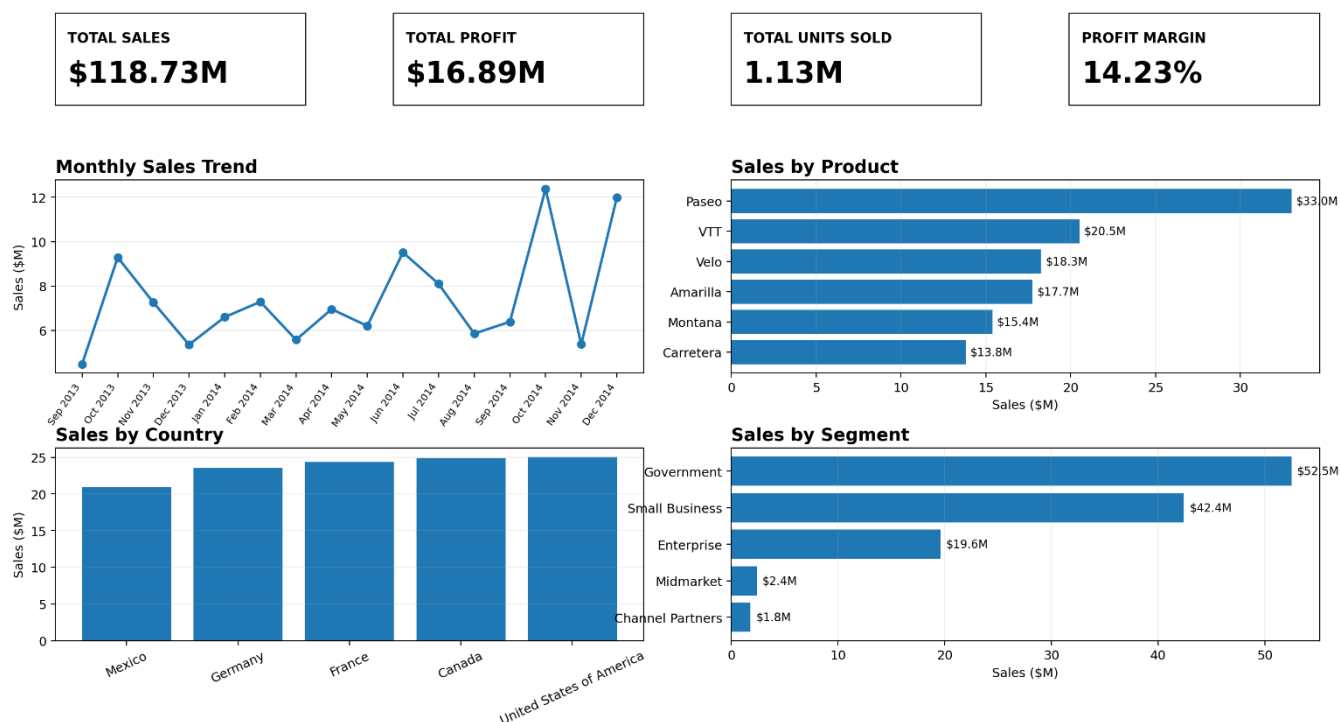
- Monitor high-sales products and segments along with their profit margins.
- Investigate the Enterprise segment to understand the reasons for its negative total profit.
- Use monthly sales trends for inventory and sales planning.
- Review discount levels because higher discounts are associated with lower profit margin in this dataset.
- Use the regression model as a supporting forecasting tool and validate predictions with future business data.

6. Power BI Dashboard

The Power BI dashboard should contain Total Sales, Total Profit, Total Units Sold and Profit Margin KPI cards; Sales by Product, Sales by Country, Sales by Segment and Monthly Sales Trend charts; and slicers for Year, Country, Product, Segment and Discount Band. The cleaned Excel workbook supplied with this project can be used as the Power BI data source.

SALES & PROFIT PERFORMANCE DASHBOARD

Business Performance Analysis • 2013–2014



7. Limitations & Next Steps

The current analysis is based on the variables and period represented in the supplied dataset. Historical relationships do not automatically establish causation, particularly for discounting and profitability. The regression model also reports performance on a 140-record test set and should be validated on future periods. Recommended next steps include adding time-based validation, monitoring model error over new periods, examining feature importance from the Random Forest model, analysing Enterprise at a more granular product/country level, and extending the Power BI dashboard with profit-focused views.

8 . Conclusion

The Week 4 capstone demonstrates a complete business analytics pipeline. The cleaned dataset contains 700 records across 16 columns, with no duplicate rows after cleaning. The business analysis establishes 118.73 million in Sales, 16.89 million in Profit and a 14.23% overall Profit Margin. Product, segment and time analysis identifies important areas for management attention, while the Random Forest model provides a strong predictive baseline with an R^2 of 0.9847 on the test set. The Power BI dashboard brings these findings together into an interactive reporting layer, allowing users to examine sales performance by product, country, segment, time and discount band. The strongest practical value comes from combining Sales with Profit and Margin so that growth and profitability can be evaluated together.

GITHUB LINK

<https://github.com/patilsaisha1404-dot/Skill-Nexis-Data-Science-Internship>